

Methanol dimer analysis report

MMML λ -dynamics TI trajectories and 2D cutoff scans

Source: artifacts/meoh_dimer_lambda_ti/ and artifacts/meoh_dimer_2d_cutoff_scan/

Generated: 2026-04-29

Contents

1. Lambda dynamics and thermodynamic integration	2
1.1. Geometry (from run summary)	2
1.2. Free-energy summary (this artifact run)	2
2. Two-dimensional cutoff scan (distance $\times \lambda$)	4
2.1. Scan parameters (from scan_2d_summary.json)	4
2.2. Total and interaction energies along distance	4
2.3. Two-dimensional energy maps	5
2.4. Force forms $-\partial E/\partial d$	6
2.5. λ -effect maps ($\lambda_{\max} - \lambda_{\min}$)	7
2.6. Cutoff-region slices	8
3. Data files	8

1. Lambda dynamics and thermodynamic integration

The trajectory workflow (`scripts/meoh_dimer_lambda_ti.py`) runs NVE sampling at fixed λ windows on the methanol dimer with the MMML hybrid calculator. Per-window samples estimate $\langle \partial U / \partial \lambda \rangle$ (via the inter-monomer energy difference between λ -on and λ -off calculators at the same geometry).

Thermodynamic integration (TI) integrates these means over λ . Multistate Bennett acceptance ratio (MBAR) recomputes full hybrid energies on stored snapshots for a coupled free-energy estimate and uncertainty.

1.1. Geometry (from run summary)

- Initial COM separation: 6 Å
- Seed COM separation: 6 Å

1.2. Free-energy summary (this artifact run)

Quantity	Value	Note
TI ΔF_{couple}	-0.4251799518133606 eV (-9.805032350772729 kcal/mol)	Binding path $\int_0^1 \langle \partial U / \partial \lambda \rangle d\lambda$
TI ΔF_{diss}	0.4251799518133606 eV	Negative of coupling path
MBAR ΔF_{couple}	$-0.3978459249637201 \pm 0.0059224126705642866$ eV ($-9.174685090995853 \pm 0.13657616635461595$ kcal/mol)	From pymbar; uncertainties in last column

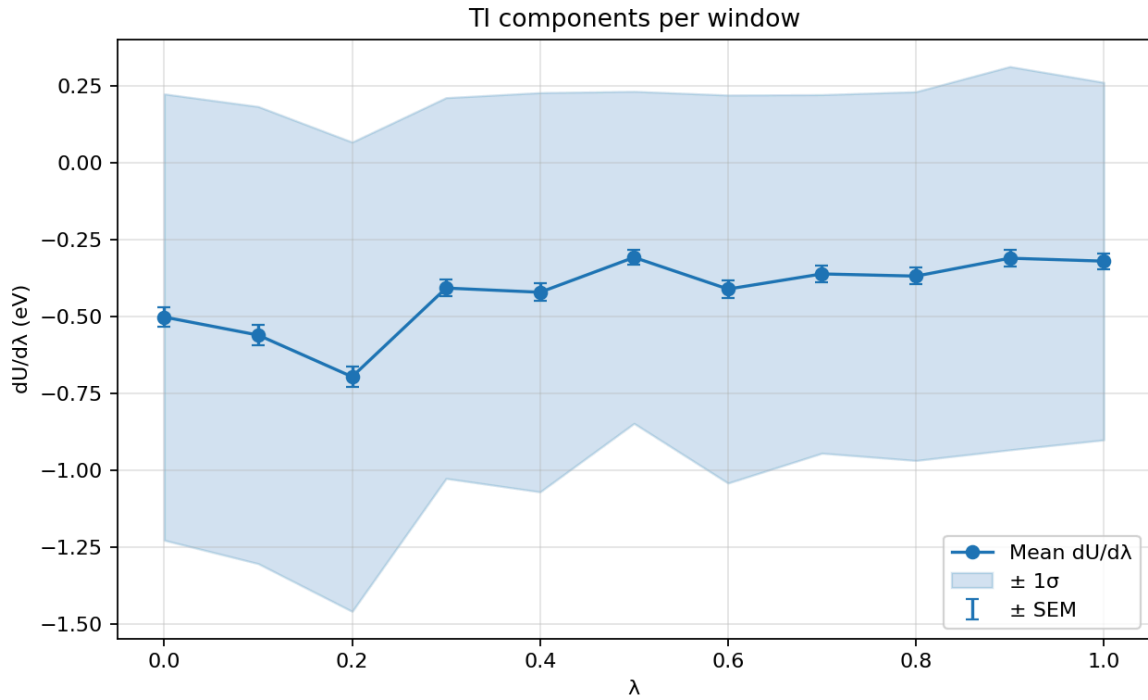


Figure 1: Mean $\partial U / \partial \lambda$ (eV) per λ window with SEM and window standard deviation.

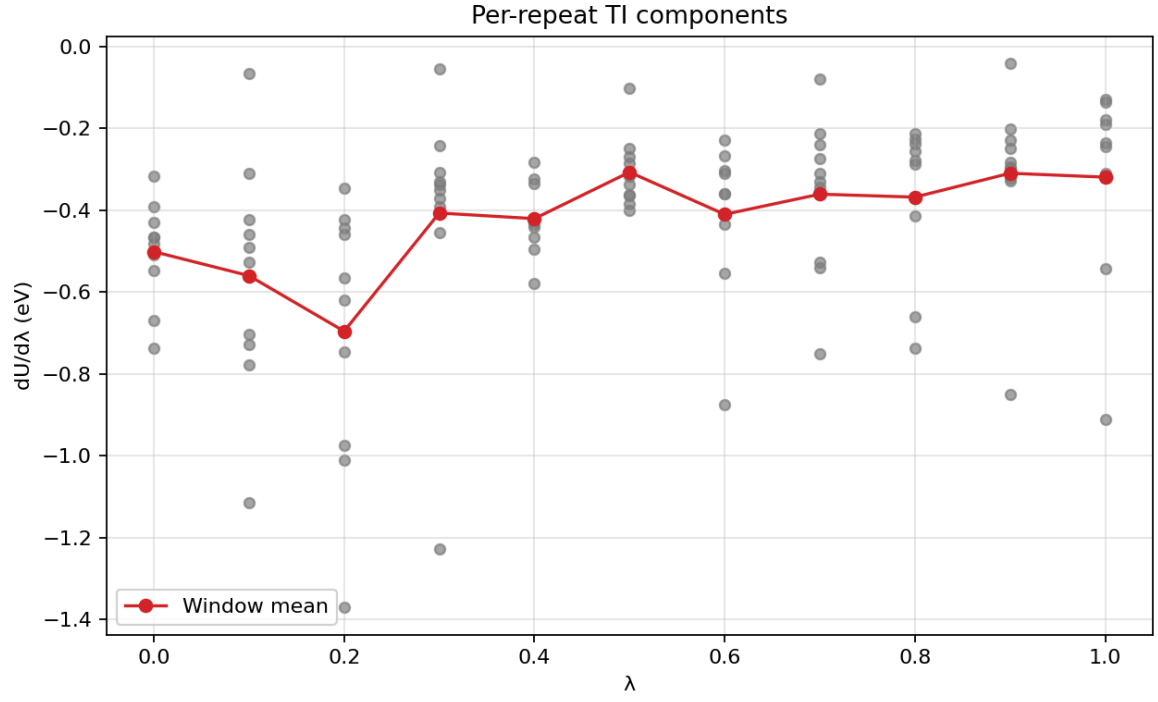


Figure 2: Repeat-level means (gray) and window mean (red): sampling spread across repeats pooled into MBAR.

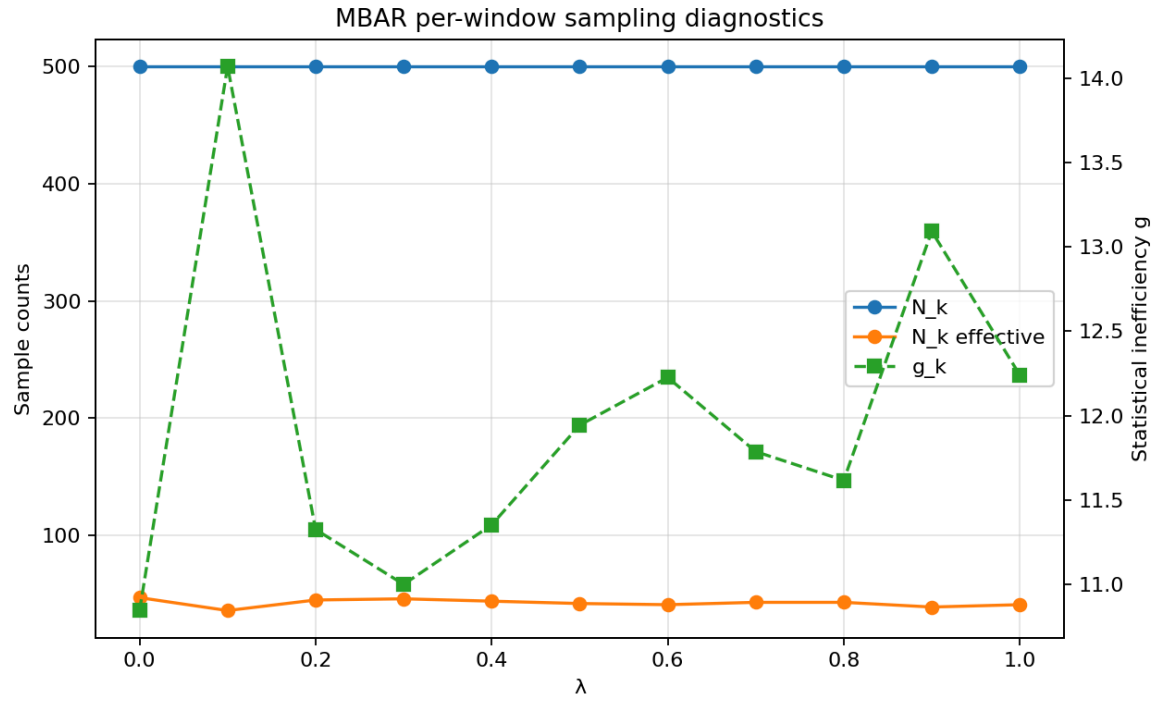


Figure 3: MBAR diagnostics: raw snapshot counts N_k , effective samples after decorrelation N_k^{eff} , and statistical inefficiency g_k per window.

2. Two-dimensional cutoff scan (distance $\times \lambda$)

The scan script (`scripts/scan_meoh_dimer_2d_cutoffs.py`) grids COM distance and alchemical λ , reporting MMML energy decomposition and force forms $-\partial E/\partial d$. Heatmaps use robust percentile color limits; vertical slices mark the complementary handoff interval between ML taper and MM switch.

2.1. Scan parameters (from `scan_2d_summary.json`)

- $d \in [0.6, 8] \text{ \AA}$, 36 distance points
- ML cutoff width: 1.5 $\text{\AA}$, MM switch end: 5 $\text{\AA}$, MM outer scale: 5 $\text{\AA}$
- Handoff interval: [3.5, 5] $\text{\AA}$

2.2. Total and interaction energies along distance

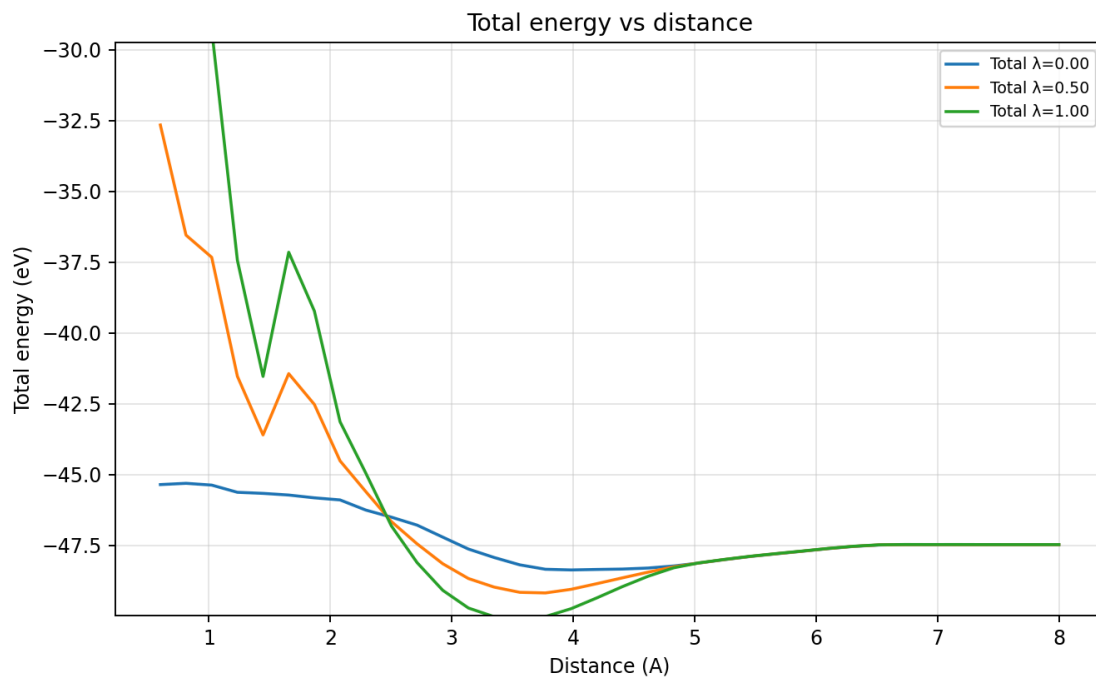


Figure 4: Total hybrid energy vs. COM distance at representative λ values.

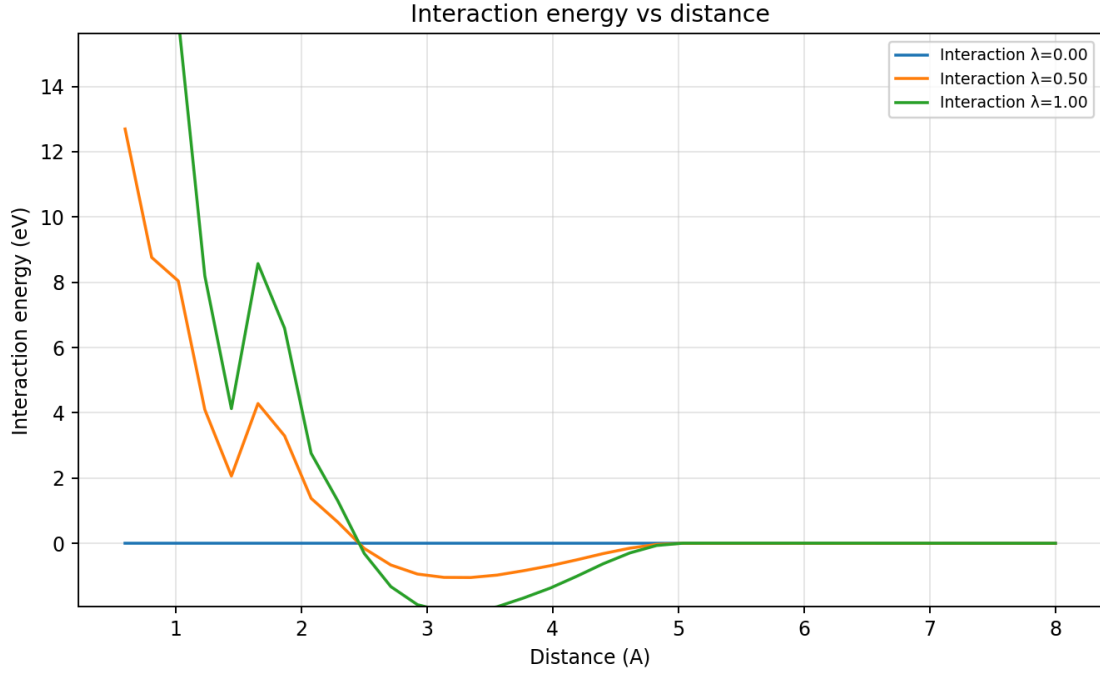


Figure 5: Interaction energy ($E_{\text{tot}} - E_{\text{internal}}$) vs. distance.

2.3. Two-dimensional energy maps

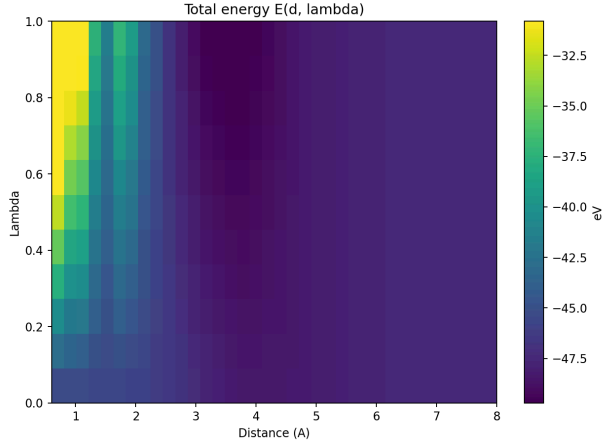


Figure 6: $E_{\text{tot}}(d, \lambda)$

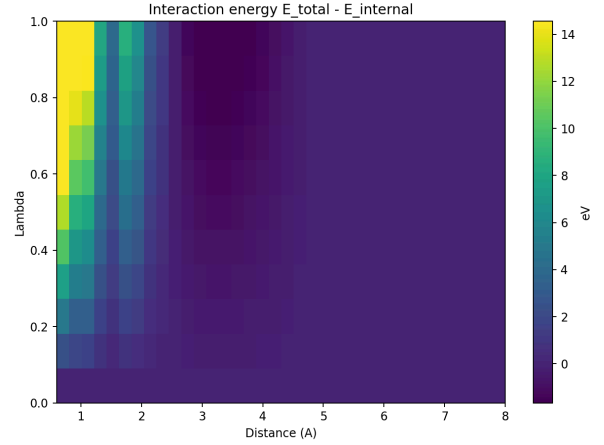


Figure 7: Interaction $E_{\text{tot}} - E_{\text{internal}}$

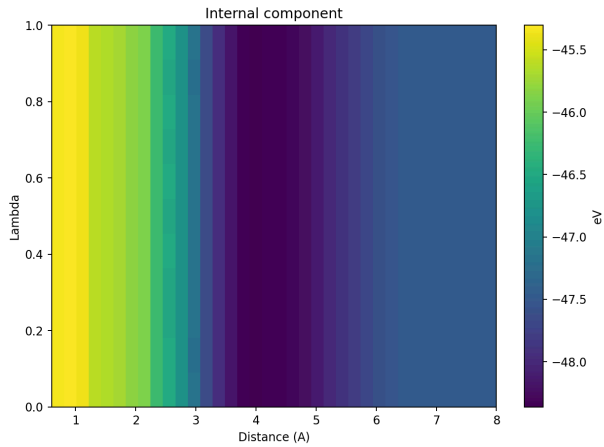


Figure 8: Internal ML monomer energy

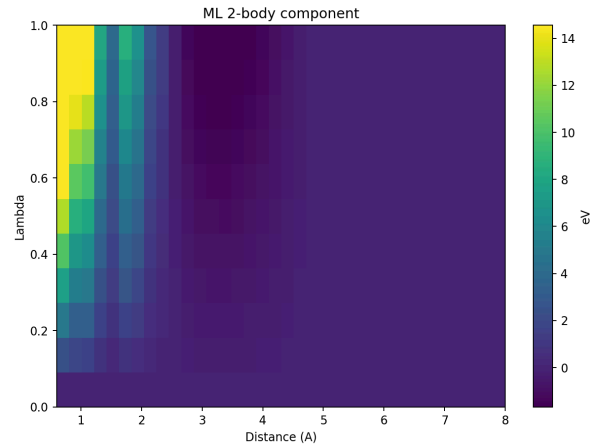


Figure 9: ML two-body (dimer) contribution

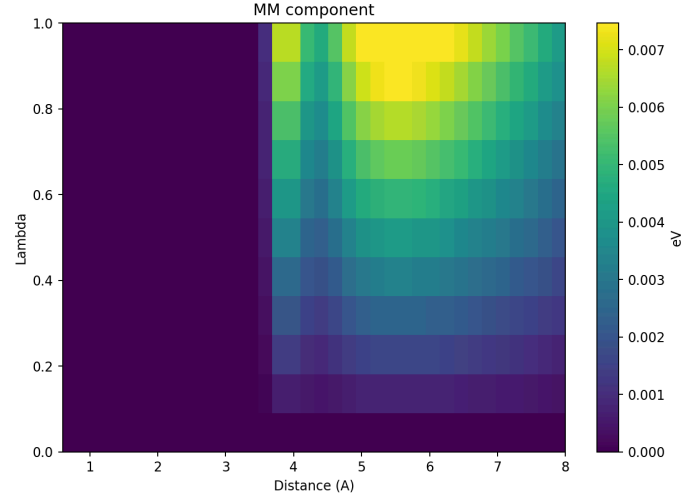


Figure 10: MM inter-monomer contribution (neighbor list excludes intra-monomer pairs).

2.4. Force forms $-\partial E/\partial d$

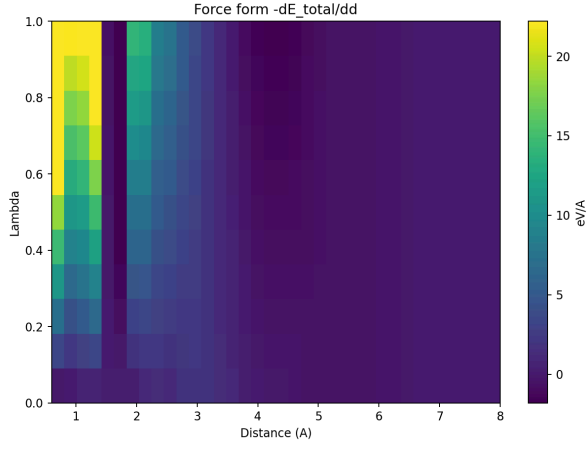


Figure 11: $-\partial E_{\text{tot}}/\partial d$

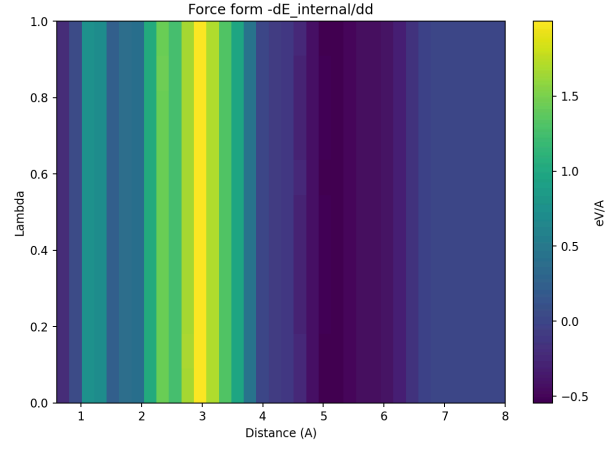


Figure 12: $-\partial E_{\text{internal}}/\partial d$

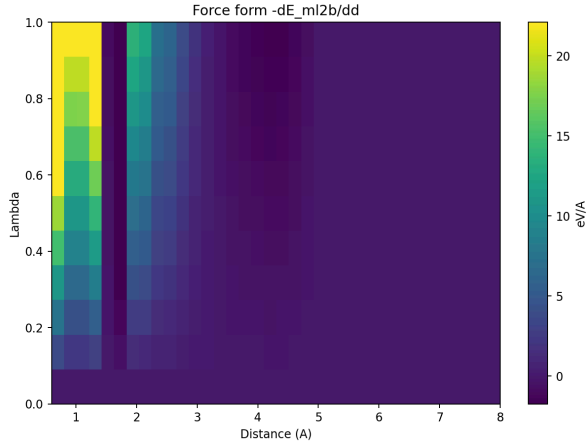


Figure 13: $-\partial E_{\text{ml2b}}/\partial d$

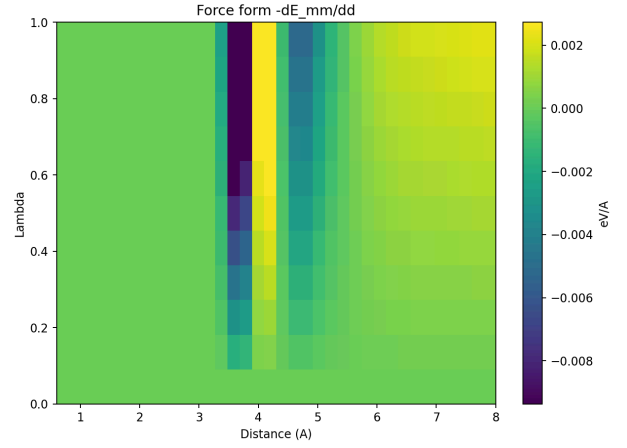


Figure 14: $-\partial E_{\text{mm}}/\partial d$

2.5. λ -effect maps ($\lambda_{\max} - \lambda_{\min}$)

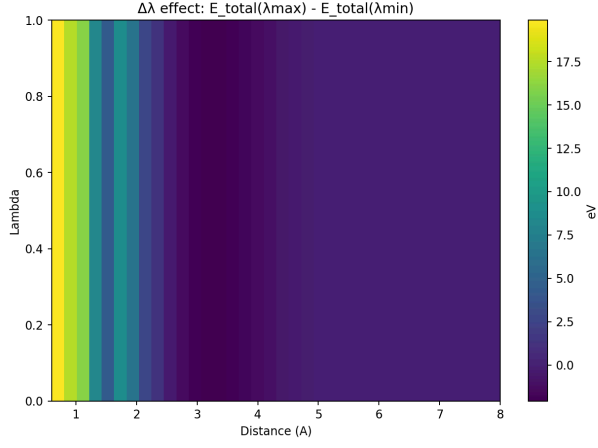


Figure 15: $\Delta_{\lambda} E_{\text{tot}}$ along d

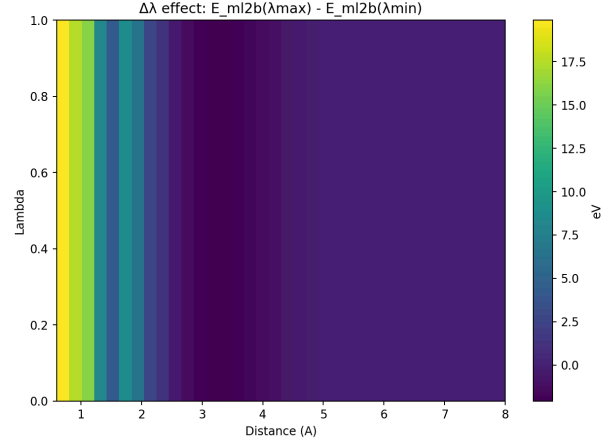


Figure 16: $\Delta_{\lambda} E_{\text{ml2b}}$

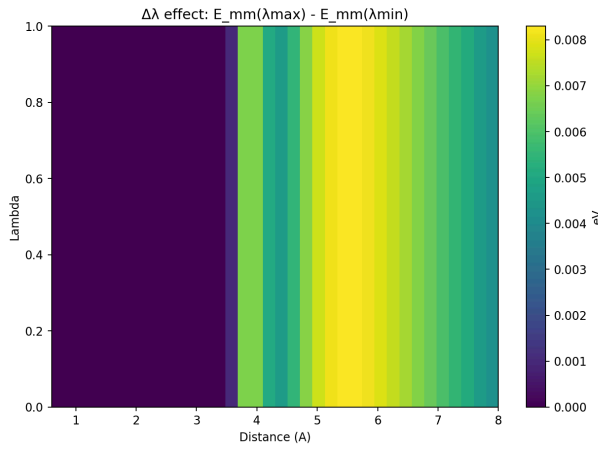


Figure 17: $\Delta_{\lambda} E_{\text{mm}}$

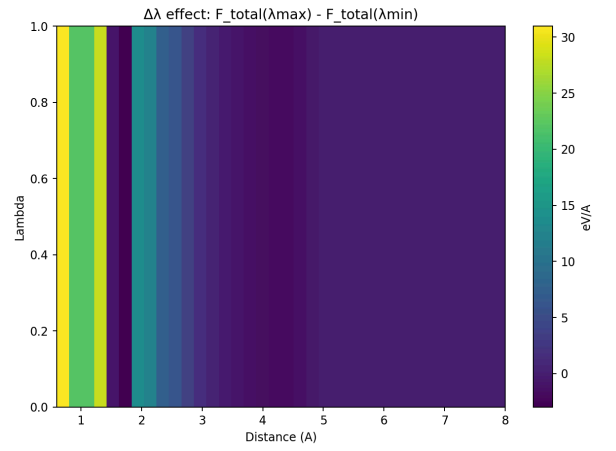


Figure 18: $\Delta_{\lambda} F_{\text{tot}}$

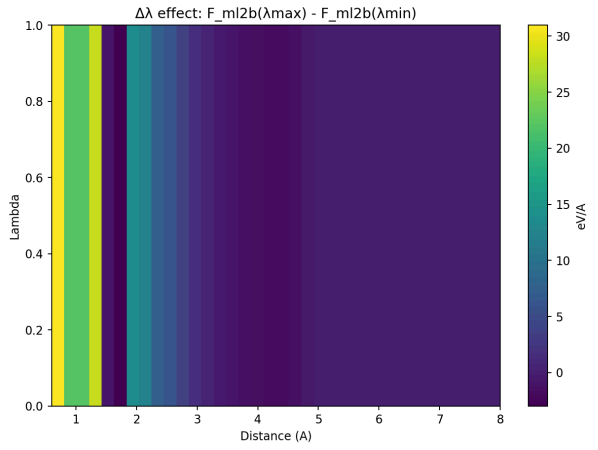


Figure 19: $\Delta_{\lambda} F_{\text{ml2b}}$

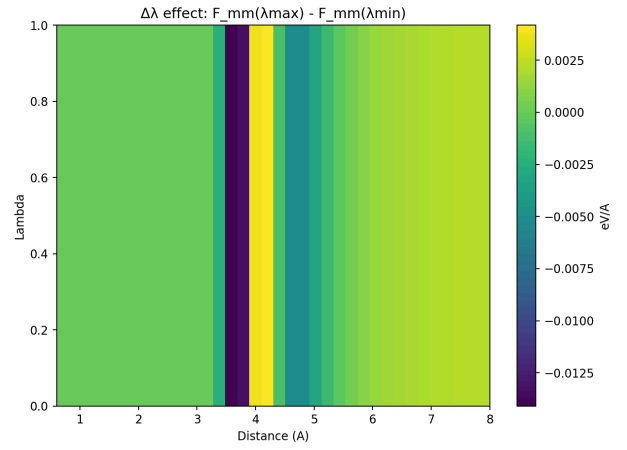


Figure 20: $\Delta_{\lambda} F_{\text{mm}}$

2.6. Cutoff-region slices

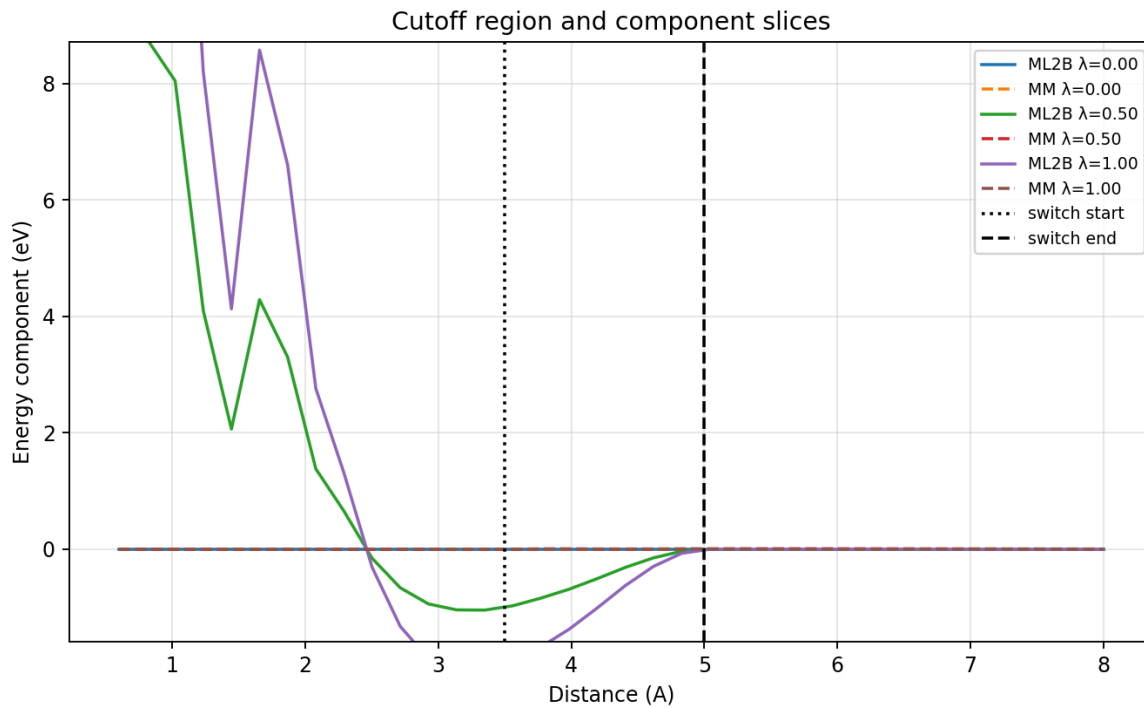


Figure 21: Energy components vs. distance at three λ values; vertical lines mark handoff start/end.

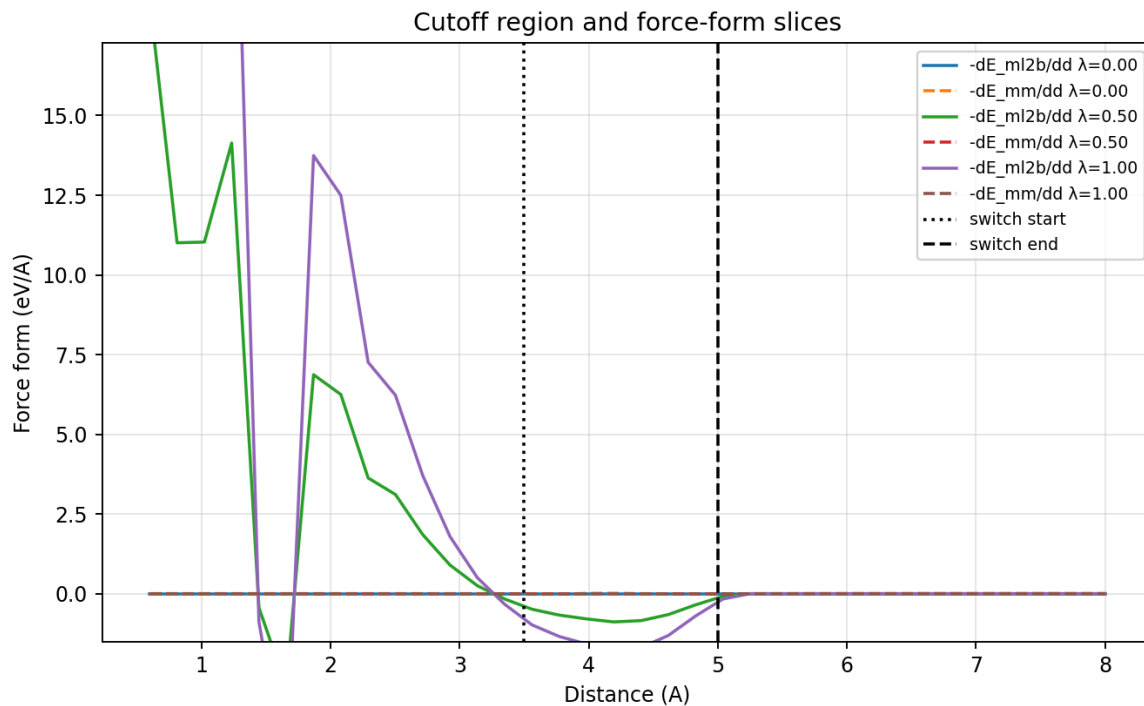


Figure 22: Force forms vs. distance with the same cutoff markers.

3. Data files

- TI: artifacts/meoh_dimer_lambda_ti/lambda_ti_summary.json, trajectories under artifacts/meoh_dimer_lambda_ti/trajectories/
- Scan: artifacts/meoh_dimer_2d_cutoff_scan/scan_2d_summary.json, scan_2d_components.csv